Revisiting In-Context Learning Dynamics Through the Lens of Antipodal Sparse Autoencoder Features

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Abstract

Despite the impressive in-context learning (ICL) capabilities of modern large language models (LLMs), our understanding of the representation-level mechanisms underlying these behaviors remains limited. As LLMs are increasingly deployed in real-world settings, closing this mechanistic gap is critical for ensuring optimal performance, safety, and controllability.

In this work, we utilize sparse autoencoders (SAEs) as an interpretability lens to investigate how LLMs perform ICL *classification*. Our analysis specifically centers on ***antipodal*** SAE features—pairs of features with approximately opposite decoder directions. We demonstrate that while these antipodal features are rare, they play an outsized role on ICL classifications. They exhibit unique characteristics, including strong inherent label selectivity and smooth activation trajectories. Moreover, we show that antipodal features provide novel insights into how few-shot demonstrations reshape internal representations. Through this lens, we expose the consistent priming effects of few-shot demonstrations primarily on the antipodal features, revealing how demonstrations drive heightened and stable label selectivity compared to zero-shot setting.[1]

1 Introduction

Mechanistic interpretability has become a central objective in contemporary LLM research, motivated by the need for safer deployment, more effective usage, and finer-grained control over model behavior (Sharkey et al., 2025; Gao et al., 2024). A particularly relevant direction is *representation engineering* (RepE) (Zou et al., 2023), which places internal representations—and the transformations between them—at the center of analysis. A representative thread within RepE is *inference-time steering*: identifying salient directions in representation space and intervening along them to modulate model behavior (Turner et al., 2023; Arditi et al., 2024). In this work, we build on this line of research to deepen our understanding of *in-context learning* (ICL), a dominant usage pattern of LLMs in which the model is prompted with a small set of demonstrations and must infer task structure and solve new instances on-the-fly, without parameter updates (Brown et al., 2020; Dong et al., 2022; Cho et al., 2024).

It is well established that adding few-shot demonstrations (FSDs) often improves performance. However, the question of *how* demonstrations produce these gains—and what precisely changes in the model's internal representations as demonstrations are incorporated—remains *not yet fully understood*. In this work, we leverage sparse autoencoders (SAEs) (Cunningham et al., 2023; Bricken et al., 2023; Lieberum et al., 2024; He et al., 2024; Karvonen et al., 2025; Gao et al., 2024) and focus on ***antipodal*** SAE features—pairs of features whose decoder directions are approximately opposite—as a principled lens for analyzing how FSDs reshape the geometry and dynamics of representation space during in-context classification. We show that this perspective yields a range of novel mechanistic insights into how demonstrations reorganize internal feature space to support stable and accurate prediction.

Our focus on antipodal features is motivated by two complementary insights:

1. **The second-slot premium in SAEs.** In an SAE with a fixed and finite dictionary, encoding a bidirectional axis with two opposing features effectively allocates two dictionary slots to a single semantic dimension. This is a meaningful capacity investment. Consequently, features that exhibit strong antipodal counterparts are natural candidates for di-

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rections that the underlying model uses both *bidirectionally* and *prominently* in representation space. Put differently, by inspecting the learned SAE features alone—without requiring any additional data—we can identify strong candidates for the model's most preferred and frequently used representational axes.

2. **Antipodal axes as a natural and efficient substrate for classification geometry.** Classification behavior often requires encoding both *evidence in favor of* a label and *evidence against* its alternatives. Antipodal feature pairs provide a natural substrate for encoding these opposing contributions along a shared representational axis. This organization makes the two signals intrinsically coupled and, to a large extent, mutually exclusive: it becomes difficult for the model to sustain high-confidence "for-L" and "against-L" signals at the same time. In this way, antipodal structure can support cleaner classification geometry by reducing internal representational conflict.

We provide empirical support for both claims. First, antipodal counterparts are rare: only $\approx 0.28\%$ of SAE features admit a strong antipodal partner under our criterion. Second, despite their scarcity, antipodal features disproportionately drive classification behavior by exhibiting the strongest label-discriminative strength. Moreover, antipodal features offer a mechanistically informative perspective on how few-shot demonstrations reshape representation space. In particular, few-shot demonstrations appear to systematically prime these axes to become more label-discriminative, to play a stronger causal role in the model's final prediction, and to exhibit greater robustness to short-timescale, token-level noise in their per-token logit traces, thereby enabling more stable predictions.

2 Sparse Autoencoders

Sparse autoencoders (SAEs) have emerged as one of the most effective tools for making the internal activations of large language models (LLMs) more interpretable to humans (Sharkey and Beren, 2022). A central challenge in understanding LLMs is that their intermediate states are high-dimensional vectors whose coordinates do not correspond to human-meaningful concepts. SAE-derived features partially bridge this gap: by learning a sparse latent code that reconstructs the model's activations, SAEs often isolate directions that behave like semantically coherent factors. As a result, SAE features are frequently described as a kind of *microscope* for probing LLM internals, enabling more targeted inspection of the computations underlying model behavior (Team, 2024; Lindsey et al., 2025).

The motivation for SAEs stems from the observation that individual neurons in LLMs typically participate in many unrelated computations and activate densely, making neuron-level interpretation difficult. SAEs address this by learning a representation in which, for any given input, only a small number of latent units are active. These sparsely active latent units are commonly referred to as *features* and can be treated as a learned dictionary of directions in representation space.

Formally, let $\mathbf{x}_i^{(\ell)} \in \mathbb{R}^d$ denote the activation vector (e.g., residual-stream state) at layer ℓ of an LLM for the i-th input. An SAE produces a reconstruction $\hat{\mathbf{x}}_i^{(\ell)}$ via an encoder–decoder factorization:

$$\hat{\mathbf{x}}_i^{(\ell)} = W_{\text{dec}}\, \sigma\Big(W_{\text{enc}}\, \mathbf{x}_i^{(\ell)} + \mathbf{b}_{\text{enc}}\Big) + \mathbf{b}_{\text{dec}}, \quad (1)$$

where W_{enc} and W_{dec} are the encoder and decoder matrices and $\mathbf{b}_{\text{enc}}, \mathbf{b}_{\text{dec}}$ are biases. The nonlinearity $\sigma(\cdot)$ enforces nonnegativity and, together with the sparsity penalty, encourages a sparse latent code. In this work we use JumpReLU (Lieberum et al., 2024), following the implementations in GemmaScope (Lieberum et al., 2024) and LlamaScope (He et al., 2024).

SAEs are trained to minimize a reconstruction objective augmented with a sparsity regularizer:

$$L_{\text{SAE}} = \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{x}_i^{(\ell)} - \hat{\mathbf{x}}_i^{(\ell)} \right\|_2^2 \; + \; \lambda\, L_{\text{sparsity}}, \quad (2)$$

where L_{sparsity} (typically an ℓ_1-style penalty on the latent code) encourages only a small number of features to activate per input, and λ controls the reconstruction–sparsity trade-off. Intuitively, the decoder columns of W_{dec} define a learned dictionary of directions in activation space, while the sparse code selects a small subset of these directions to explain each activation vector. This sparse, dictionary-based view is what makes SAE features a natural unit of analysis for mechanistic interpretability.

3 Revisiting In-Context Learning Dynamics Through Antipodal Features

3.1 Overview

We study the following central question:

"Can sparse autoencoders provide novel insights into how LLMs perform classification, and how few-shot demonstrations reshape their internal representations?"

Sparse autoencoders (SAEs) are increasingly used to interpret LLM representations, but it remains unclear what their features reveal mechanistically and how they relate to task computation (Deng et al., 2025; Cho and Hockenmaier, 2025). In this work, we show that richer insight can be obtained by focusing on *antipodal* SAE features, formally defined in Section 3.2. Using these features as a diagnostic lens, we show that they play an outsized role in classification and provide new evidence that few-shot demonstrations (FSDs) reshape representation space by *priming* certain features to become more selective and more central to the model's decision process.

Our main empirical finding is that, in standard nonnegative-code SAEs, a very small fraction of features (fewer than 0.5%) have clear antipodal counterparts. Despite their rarity, these features exhibit unusually strong label selectivity, greater robustness to short-timescale token-level fluctuations in their logit traces, and substantial contextual flexibility: under FSDs, their selectivity sharpens and their functional role can shift across tasks (e.g., from a SPORTS detector in AGNews to a NUMERIC detector in TREC).

We use `Gemma-2-9B-IT` (Team et al., 2024) and `Llama3.1-8B-Instruct` as our primary testbeds. Our experiments mainly use the Yahoo Answers Topics dataset, with AG News as a supplementary validation benchmark (Zhang et al., 2015).

3.2 Definition of Antipodal SAE Features

SAE Setup and Notation Let $h_t^{(\ell)} \in \mathbb{R}^d$ denote the residual-stream representation at layer ℓ and token position t. We analyze a pretrained sparse autoencoder (SAE) that maps $h_t^{(\ell)}$ to a sparse nonnegative code $a_t \in \mathbb{R}^m_{\geq 0}$ and reconstructs it via

$$a_t = \phi\Big(W_{\text{enc}} h_t^{(\ell)} + b_{\text{enc}}\Big)\,,\ \hat{h}_t^{(\ell)} = W_{\text{dec}} a_t + b_{\text{dec}}$$

We refer to the f-th decoder column,

$$d_f := W_{\text{dec}}[:, f] \in \mathbb{R}^d$$

as feature f. Intuitively, SAE features are learned dictionary directions in the model's representation space, and the sparse code selects a small subset of these directions for each token representation.

Definition of Antipodal Feature Pairs in SAEs Nonnegative-code SAEs cannot represent a signed scalar coefficient along a direction using a single feature activation, since $a_{t,f} \geq 0$. When the underlying model representation varies *bidirectionally* along a latent axis—as suggested by recent work (Zhu et al., 2025)[Please Include Two More Famous References Here]—a natural SAE solution is to allocate two features whose decoder directions are approximately antipodal. Empirically, we find that trained SAEs contain a compact subset of such $\pm$ pairs. We formalize this phenomenon as *antipodal pairing*.

Definition 1 (Antipodal Feature Pairs)**.** Fix a threshold $\tau \in (0, 1)$ (we use $\tau = 0.95$ unless stated otherwise). For a feature f, define its most antipodal partner:

$$g^\star(f) := \arg\min_{g \neq f} \cos(d_f, d_g) \tag{3}$$

We say that feature f is *antipodal-paired* if

$$\cos\big(d_f, d_{g^\star(f)}\big) \ \leq\ -\tau. \tag{4}$$

In this case, we call $g^\star(f)$ the *antipodal counterpart* of f, and we refer to $(f, g^\star(f))$ as a *antipodal pair*. Throughout, we use *antipodal feature* as shorthand for a feature that has a antipodal counterpart in the SAE.

Why Focus on Antipodal Features?

We provide our two key insights behind why the antipodal features could be a good lens for studying representation dynamics in LLMs.

Key Intuition #1: The *"Second-Slot" Premium*

Our first intuition is that, in the standard SAE setting with *nonnegative* activations, learning an explicit antipodal partner is not costless; rather, it reflects a meaningful allocation of scarce dictionary capacity. An SAE has a fixed dictionary width m (e.g., $m = 16{,}384$), so dedicating an additional feature to represent a direction close to $-d_f$ necessarily comes at the expense of some other potential dictionary atom. In this sense, an antipodal partner occupies *premium real estate* in a finite dictionary: the optimizer is likely to "pay" for this second slot only when the underlying model traverses that latent axis in *both* directions, and does so *saliently enough* to justify the extra capacity.

Key Intuition #2: Antipodal Axes as a Natural Substrate for Classification Our second intuition is that antipodal features provide a particularly natural substrate for classification-relevant computation. Classification often involves not only accumulating *evidence for* a candidate label, but also accumulating *evidence against* competing labels. Encoding these opposing signals along a shared antipodal axis yields a coherent margin geometry: strong "for-L" evidence corresponds to movement in one direction, while strong "against-L" evidence corresponds to movement in the opposite direction. This coupling makes it difficult for the model to simultaneously sustain high-confidence "for-L" and "against-L" signals, thereby reducing internal inconsistency and supporting cleaner decision boundaries.

3.3 Unique Properties of Antipodal Features

(1) Existence and Compactness

We begin by quantifying how frequently antipodal features arise in standard nonnegative-code SAEs. For this analysis, we use `Gemma2-9B-IT` together with its corresponding JumpReLU-based SAEs (Lieberum et al., 2024). Applying our definition of antipodal pairing, we find that only 52 features (i.e., 26 pairs), which is approximately 0.12% of all SAE features, admit a strong antipodal counterpart. Overall distribution of features are shown in Figure 1a.

This rarity is itself informative. It is consistent with the *second-slot premium* intuition: under a fixed decoder width, allocating an additional dictionary slot to represent the opposite polarity of an existing direction is a meaningful capacity investment. More generally, it may also reflect a property of the underlying base model itself—namely, that only a small number of latent axes are traversed *saliently* in both directions. These two explanations are not mutually exclusive.

Importantly, as we show below, antipodal features exhibit several distinctive and desirable properties. From this perspective, their compactness is not a limitation but an advantage. For researchers seeking an efficient starting point for mechanistic analysis on a new task, antipodal features provide a small, high-yield set of candidates that is especially likely to reveal meaningful structure. A further practical advantage is that identifying this feature set requires no additional task-specific data: one can inspect a trained SAE directly, compute pairwise cosine similarities among decoder directions, and extract the antipodal subset. In this sense, antipodal features offer a compact and data-efficient candidate set of representational axes that the model appears to use particularly saliently.

(2) Strong Inherent Label Discrimination

We further show that antipodal features exhibit *strong inherent label-discriminative behavior*. Concretely, even in the zero-shot setting—where the model receives only a task instruction followed immediately by the test query—the features with the highest label-discrimination scores (defined below) are disproportionately enriched within the antipodal subset. This result, summarized in Figure **??**(a) (with full results in Appendix 5), supports our second key intuition: *antipodal axes form a core classification geometry that the model naturally exploits.*

Formally, let $\mathcal{Y} = \{1, \ldots, C\}$ denote the set of C labels, and let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$ be an evaluation set. For each example i and SAE feature f, we define a scalar feature-logit summary $s_{i,f}$ as the pre-activation value of feature f at the final token of the test query, i.e., the position used to predict the query label.

We then compute the class-conditional means:

$$\mu_f(y) \;:=\; \mathbb{E}_{i:\, y_i = y}\big[s_{i,f}\big], \qquad y \in \mathcal{Y}. \tag{5}$$

Our primary discrimination statistic is the one-vs.-rest *margin* for each label:

$$\Delta_f(y) \;:=\; \mu_f(y) - \frac{1}{C-1} \sum_{y' \in \mathcal{Y} \setminus \{y\}} \mu_f(y'). \tag{6}$$

Intuitively, $\Delta_f(y)$ measures how strongly feature f favors label y relative to the competing labels.

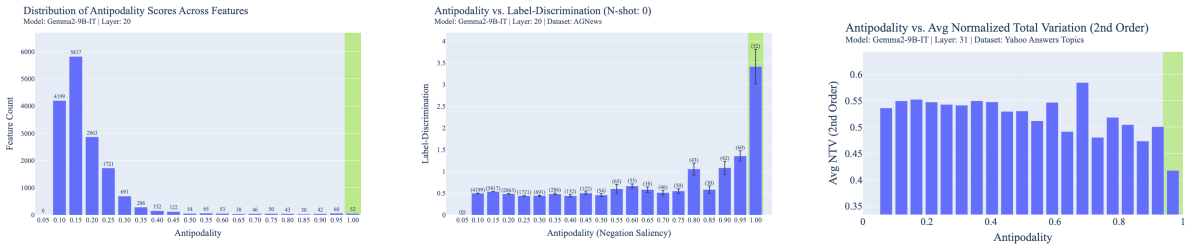
Finally, we define the *label-discrimination strength* of feature f as

$$\mathrm{LD}(f) \;:=\; \max_{y \in \mathcal{Y}} \big|\Delta_f(y)\big|, \tag{7}$$

and optionally assign the feature to its most strongly associated label:

$$y_f^{\star} \;:=\; \arg\max_{y \in \mathcal{Y}} \big|\Delta_f(y)\big|. \tag{8}$$

We refer to features with large $\mathrm{LD}(f)$ as *label-discriminative* features. In Figure 1b, we compute $\mathrm{LD}(f)$ under the instruction-only zero-shot condition and find that the most label-discriminative features are substantially enriched among antipodal features. This result indicates that the model already leverages antipodal axes to encode strong class-separating structure, even before any few-shot demonstrations are introduced.



(a) Antipodal features are rare within the dictionary (0.318%).

(b) Antipodal features are intrinsically (zero-shot) label-discriminative (LD).

(c) Antipodal features are inherently smoother.

Figure 1: **Unique characteristics of antipodal features and their primings by few-shot demonstrations (FSDs).** The green region denotes the antipodal-feature subset.

(3) Smoother Activation Trajectories

We next show that antipodal features exhibit smoother token-level activation trajectories over the test query tokens. Relative to non-antipodal features, their feature logits display less short-timescale irregularity—that is, fewer abrupt token-to-token reversals and smaller local oscillations. We regard this as a meaningful diagnostic signal: features with highly jagged trajectories are more likely to reflect brittle, surface-level responses driven by lexical or formatting cues, whereas features with lower local roughness are more consistent with stable, higher-level semantic evidence. Because our goal is to identify features involved in classification-relevant computation, we treat short-timescale smoothness as an informative indicator of mechanistic relevance.

Normalized Total Variation (NTV) To quantify short-timescale roughness, we use scale-normalized first- and second-order total variation. We begin by defining the trajectory amplitude:

$$A^{(i,f)} := \max_{1\le t\le G} z_t^{(i,f)} - \min_{1\le t\le G} z_t^{(i,f)}, \quad (9)$$

which measures the overall range of the trajectory for feature f on example i which consists of G tokens. We then define the normalized total variation statistics:

$$\mathrm{NTV}_1^{(i,f)} := \frac{\frac{1}{G-1}\sum_{t=1}^{G-1}\left|z_{t+1}^{(i,f)} - z_t^{(i,f)}\right|}{A^{(i,f)} + \varepsilon} \quad (10)$$

$$\mathrm{NTV}_2^{(i,f)} := \frac{\frac{1}{G-2}\sum_{t=1}^{G-2}\left|z_{t+2}^{(i,f)} - 2z_{t+1}^{(i,f)} + z_t^{(i,f)}\right|}{A^{(i,f)} + \varepsilon} \quad (11)$$

where $\varepsilon > 0$ is a small constant for numerical stability. Here, $A^{(i,f)}$ serves as a scale normalizer, so these quantities can be interpreted as measures of *relative local roughness*. NTV_1 captures the average token-to-token step size, whereas NTV_2 captures changes in slope and is therefore especially sensitive to rapid up–down oscillations. Accordingly, we treat NTV_2 as our primary smoothness metric and report NTV_1 as a complementary robustness check. We aggregate these quantities across examples to obtain feature-level summaries by averaging over i.

The results are reported in Figure 1c (results on other models and tasks are provided in Appendix X). We find that antipodal features consistently exhibit lower scale-normalized roughness (NTV1/NTV2) than non-antipodal features, indicating that their activation traces are smoother and less locally noisy at the token level. This pattern supports the view that antipodal features encode more stable, potentially task-relevant, evidence across the test query.

3.4 Antipodal Features Reveal How Few-Shot Demonstrations Reshape Internal Representation

While it is well known that few-shot demonstrations (FSDs) often improve downstream performance, our representation-level understanding of *how* they reshape internal geometry and computation remains limited. We show that antipodal features provide a particularly informative lens on this process. Across a range of shot counts, we observe two consistent *priming* effects:

1. FSDs strengthen the *label discrimination* of antipodal features, as measured by increased discrimination strength $\mathrm{LD}(f)$ (Equation (7)).

2. FSDs make antipodal feature traces *smoother*, reducing short-timescale jitter as captured by normalized variation metrics (i.e., Eqs. (10)–(11)).

3. FSDs increase the *causal efficacy* of antipodal features: the same intervention along an antipodal direction produces substantially larger shifts in output probabilities under few-shot prompting than in the zero-shot setting. In other words, FSDs not only make these features more selective, but also make the model more causally sensitive to them when forming its final prediction.

We elaborate on these findings below.

(1) FSDs Strengthen the Label Discrimination of Antipodal Features

To quantify this effect, we track $\mathrm{LD}(f)$ for each feature across shot counts $N \in \{0, 8, 16, 24\}$ and report the aggregate trends in Figure 2a (full results are provided in Appendix 5). We find that increasing the number of demonstrations systematically increases $\mathrm{LD}(f)$, indicating that FSDs sharpen the class-discriminative behavior of antipodal features. In other words, few-shot demonstrations amplify the margin-encoding component of these features.

Moreover, this strengthening effect is substantially more pronounced for antipodal features than for non-antipodal features. This suggests that FSDs do not uniformly enhance all SAE features; rather, they preferentially prime a compact subset of antipodal features that appears to play a central role in the model's classification mechanism. Put differently, antipodal features are already intrinsically more label-selective in the zero-shot setting (Figure 1b), and they become even more discriminative as few-shot demonstrations are introduced (Figure 2a).

(2) FSDs Make Antipodal Feature Trajectories Smoother

A second priming effect is that FSDs make antipodal feature trajectories *more stable* across tokens. We quantify this effect using normalized total variation, treating $\mathrm{NTV}_2^{(f)}$ as our primary smoothness metric and reporting $\mathrm{NTV}_1^{(f)}$ as a complementary robustness check (Equations (10)–(11)). As the number of demonstrations N increases, both metrics decrease substantially: $\mathrm{NTV}_1^{(f)}$ falls from 11.29 to 9.25, while $\mathrm{NTV}_2^{(f)}$ falls from 9.00 to 6.00, corresponding to average reductions of approximately 18.1% and 33.3%, respectively. This consistent decline indicates that FSDs do not merely strengthen discriminative evidence; they also *stabilize* the temporal logit traces of antipodal features by suppressing short-timescale fluctuations. The resulting representation is therefore not only more discriminative, but also smoother and more robust at the token level.

(3) FSDs Increase Not Only Distributional Selectivity, but Also the Causal Influence of Antipodal Features

The preceding analysis shows that FSDs amplify antipodal features and sharpen their label selectivity in a distributional sense. A stronger and more consequential question is whether these same features also become more *causally influential* in the model's decision process. Establishing this connection would substantially deepen our understanding of how few-shot prompting operates: it would suggest that FSDs do not merely alter feature statistics passively, but instead increase the extent to which the model *functionally reads out* antipodal features when producing its final prediction. Our results support precisely this interpretation. Once FSDs are introduced, highly label-selective antipodal features become substantially more effective at steering the model toward their associated labels.

We operationalize causal influence using an activation-steering intervention. For each label l, let

$$\mathcal{T}_l^{(k)} = \arg\operatorname*{topk}_j \ \mathrm{LD}_l(j)$$

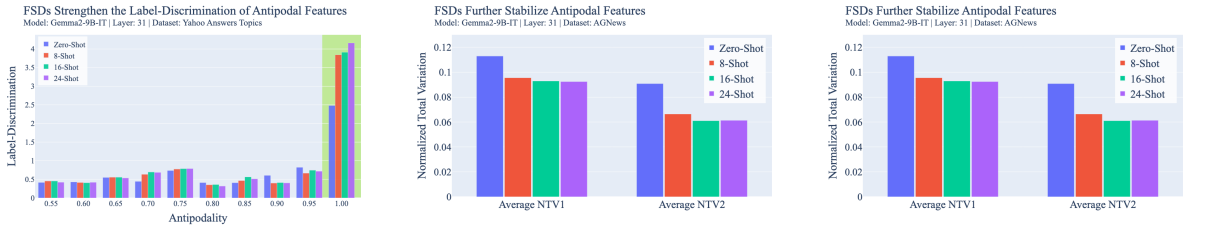
be the set of the top-k antipodal features most selective for label l. For an evaluation example x, let $p^{(N)}(x)$ denote its prompt instantiation under the N-shot setting, and let $Q(x)$ denote the token positions corresponding to the test query. Let $h_t\big(p^{(N)}(x)\big) \in \mathbb{R}^d$ be the hidden state at token position t. Steering feature j with strength α corresponds to the intervention

$$\tilde{h}_t^{(j,\alpha)}\Big(p^{(N)}(x)\Big) = \begin{cases} h_t\Big(p^{(N)}(x)\Big) + \alpha d_j, & t \in Q(x) \\ h_t\Big(p^{(N)}(x)\Big), & t \notin Q(x) \end{cases}$$

Let $p_\theta^{(N)}(y \mid x)$ denote the model's output distribution without intervention under the N-shot prompt, and let $p_\theta^{(N;j,\alpha)}(y \mid x)$ denote the output distribution obtained after replacing $\{h_t\}_{t\in Q(x)}$ with $\{\tilde{h}_t^{(j,\alpha)}\}_{t\in Q(x)}$ and continuing the forward pass. We focus on the induced change in the target-label probability:

$$\Delta_{j,l}^{(N)}(x;\alpha) = p_\theta^{(N;j,\alpha)}(y = l \mid x) - p_\theta^{(N)}(y = l \mid x)$$

This quantity measures how much steering along feature j increases the model's tendency to predict



(a) antipodal features are preferentially primed by FSDs (LD).

(b) FSDs Further Stabilize Antipodal Features.

(c) FSDs Further Stabilize Antipodal Features.

Figure 2: **Title...** (a) ... (b) This discriminative signal is further amplified as the number of few-shot demonstrations (FSDs) increases. (c) FSDs also stabilize antipodal features' per-token logit traces, reducing short-timescale jitter and increasing robustness to superficial token-level noise.

label l. We then average over a fixed evaluation set $\mathcal{D}$ containing 300 test queries per label:

$$\mathrm{CE}_{j,l}^{(N)}(\alpha) = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} \Delta_{j,l}^{(N)}(x; \alpha).$$

We compare these quantities between the zero-shot condition ($N = 0$) and the N-shot condition to assess how FSDs affect the causal efficacy of antipodal features.

Empirically, for the majority of label-selective antipodal features, we observe

$$\mathrm{CE}_{j,l}^{(N)}(\alpha) \;>\; \mathrm{CE}_{j,l}^{(0)}(\alpha)$$

across labels. This shows that FSDs do not merely sharpen label selectivity at the level of feature distributions; they also increase the *causal leverage* of those features over the model's final decision. In practical terms, the same intervention along an antipodal direction can induce a dramatically larger shift in label probability under few-shot prompting than under zero-shot prompting. Put differently, FSDs make these features not only better statistical markers of a label, but also more strongly and more effectively *read out* by the model when forming its final prediction. Illustrative examples are shown in Figure X, and aggregate statistics are reported in Table Y.

X and Y to be added very soon.

3.5 Promising Future Work

Our analysis also reveals an intriguing pattern in the distribution of antipodal features. In particular, the label-discrimination measure defined in Equation 7 is markedly uneven across labels: some labels are associated with many strongly discriminative antipodal features, whereas others are associated with relatively few. This observation raises a natural question for future work: can heterogeneity in the distribution of label-discriminative antipodal features (Figure 3) provide a new lens for understanding label bias in LLMs?

This question is especially interesting because label bias—the tendency of a model to favor certain output labels irrespective of the input content—has already been identified as a persistent issue in prompted LLM classification (Zhao et al., 2021; Fei et al., 2023). We hypothesize that the pronounced cross-label imbalance in antipodal-feature availability may help explain why some labels are systematically preferred over others. If so, antipodal-feature geometry could offer a mechanistic perspective on label bias that complements existing behavioral analyses. We leave this connection to future work.

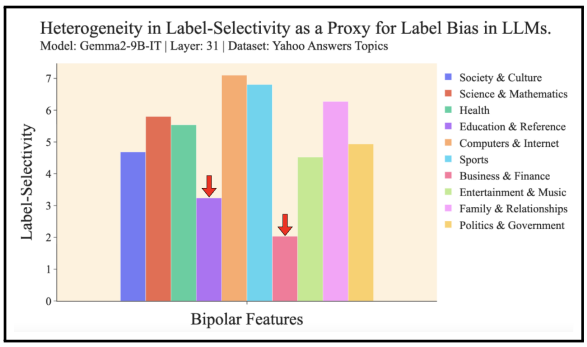


Figure 3: **Heterogeneity in Label-Selectivity as a Potential Proxy for Label Bias in LLMs.** We posit that this phenomenon serves as a novel proxy for bias, warranting further investigation in future studies.

4 Conclusion

This paper introduces *antipodal* SAE features as a primary interpretability lens for analyzing the internal dynamics of in-context learning (ICL) in *classification* settings. We show that, despite being rare, antipodal features exhibit a set of distinctive and mechanistically informative properties: (1) large

yet smooth temporal logit swings across tokens, indicating strong but low-jitter evidence traces; (2) pronounced contextual adaptivity via adaptive logit re-centering under changes in prompting context; and (3) strong inherent label-discrimination, even in instruction-only (zero-shot) regimes.

We further demonstrate that few-shot demonstrations systematically *prime* these features: increasing the number of demonstrations (i) strengthens their label-discriminative signal and (ii) further suppresses short-timescale jitter, yielding more stable per-token trajectories. Together, these findings highlight the interpretability value of antipodal features as a compact set of bidirectional axes that the model reuses as core classification geometry, and they surface important caveats for SAE-based analysis—most notably, that activation-only views can miss decision-relevant dynamics when logits undergo prompt-dependent re-centering. Overall, our results position antipodal SAE features as a high-leverage substrate for mechanistic study of ICL classification that open up multiple avenues for future work.

5 Limitations

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A Additional Results on More Models and Datasets

(1) Large and Smooth Swings We first show that antipodal features exhibit a distinctive combination of behaviors: their logits undergo *large swings* across the test query, while remaining *remarkably smooth* at short timescales. Concretely, compared to non-antipodal features, antipodal features display substantially less token-level "jitter"—i.e., fewer abrupt up–down fluctuations over one to a few tokens. We view this stability as a meaningful diagnostic signal. Features whose responses fluctuate sharply at the token scale often track lexical or superficial cues (e.g., punctuation, formatting, or function words), whereas features that vary smoothly over longer spans—while still exhibiting large amplitudes—are more likely to reflect higher-level, semantic evidence.

This distinction is particularly relevant for our goal: identifying core features that participate in the model's classification mechanism. Classification-relevant features are expected to represent persistent, concept-level evidence rather than transient lexical artifacts. Accordingly, we quantify two complementary properties of per-token traces: (i) *amplitude*, capturing the magnitude of semantic evidence accumulation, and (ii) *smoothness*, capturing robustness to short-timescale perturbations.

Amplitude For each example i and SAE feature f, let

$$z_{1:G}^{(i,f)} \in \mathbb{R}^G \tag{12}$$

denote the per-token feature logit (or activation) trace over the *test query* consisting of G tokens. We define the amplitude of a trace using a robust peak-to-peak statistic:

$$A^{(i,f)} := Q_{0.95}\big(z_{1:G}^{(i,f)}\big) - Q_{0.05}\big(z_{1:G}^{(i,f)}\big), \tag{13}$$

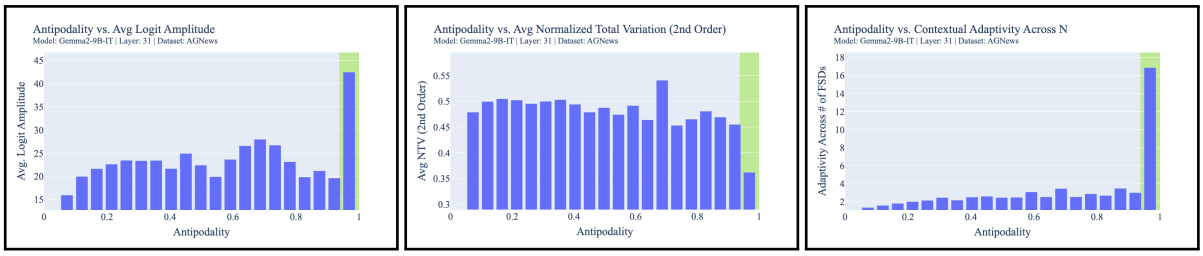
where $Q_p(\cdot)$ denotes the p-quantile over token positions $t \in \{1, \ldots, G\}$.

(2) Adaptive Logit Re-Centering A second distinctive property of antipodal features is their strong *contextual adaptivity*: they exhibit pronounced *logit re-centering* as the prompting context changes. Concretely, even when evaluated on the *same* fixed development set, the feature's average logit can shift substantially as a function of the number of few-shot demonstrations (FSDs) N.

To quantify this effect, let $s_{i,f}(N)$ denote the scalar summary of feature f on example i under an N-shot prompt computed as:

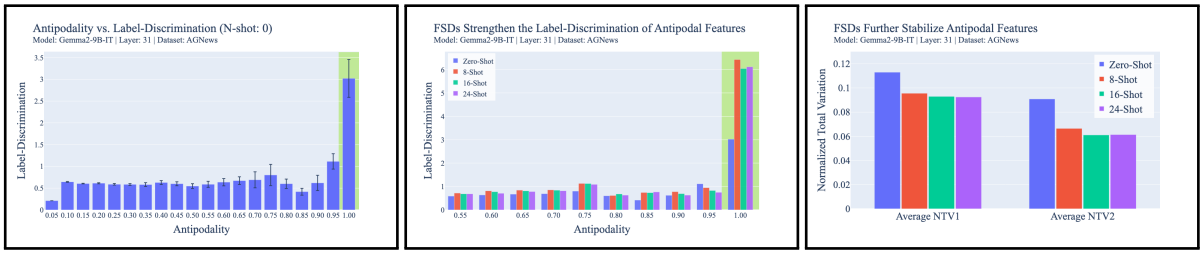
$$s_{i,f}(N) := \frac{1}{|T_i|} \sum_{t \in T_i} z_t^{(i,f)}(N) \tag{14}$$

where $z_t^{(i,f)}(N)$ is the pre-nonlinearity feature logit at token t, and T_i denotes the set of final-query



(a) Amplitude (b) Normalized Total Variation (c) Contextual Adaptivity

Figure 4: **Additional Results on Distinct Characteristics of Antipodal Features (Gemma2-9B-IT, AGNews).**



(a) Inherent Label-Discrimination (Zero-Shot) (b) FSDs Prime Antipodal Features to Become Even More Discriminative. (c) FSDs Further Stabilize Antipodal Features

Figure 5: **Additional Results on Label Discrimination and Few-Shot Priming of Antipodal Features (Gemma2-9B-IT, AGNews).**

token positions for example i. We choose T_i to be the last 15 tokens of the *final query*.

We then define the dev-set average logit as

$$\bar{s}_f(N) \;:=\; \mathbb{E}_{i\sim\mathcal{D}}\big[s_{i,f}(N)\big], \tag{15}$$

and measure the magnitude of re-centering across a set of shot counts $\mathcal{N} = \{0, 8, 16, 24\}$) using the total variation across N:

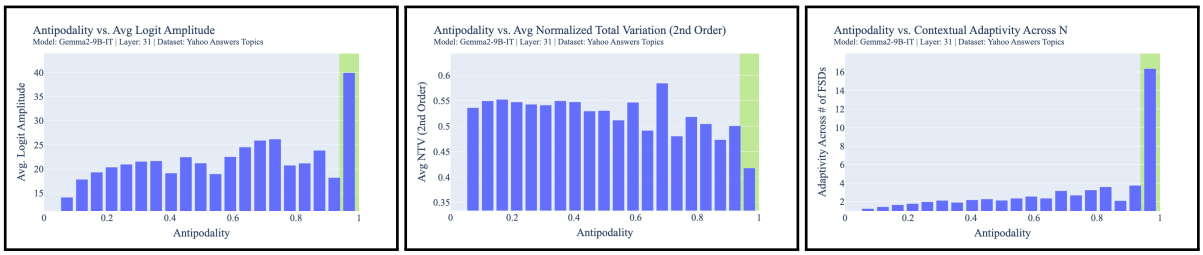
$$\mathrm{TV}_N(f) \;:=\; \sum_{k=1}^{|\mathcal{N}|-1} \big|\bar{s}_f(N_{k+1}) - \bar{s}_f(N_k)\big|, \tag{16}$$

$$\big(\mathcal{N} = \{N_1 < \cdots < N_{|\mathcal{N}|}\}\big) \tag{17}$$

Intuitively, $\mathrm{TV}_N(f)$ captures how actively the feature's *baseline* moves as demonstrations are added, even though the underlying evaluation examples are held fixed. We report this metric in Figure 6 (c) (and additional results in Appendix X) to compare the degree of shot-conditioned re-centering between antipodal and non-antipodal features. As illustrated, antipodal features exhibit significantly higher logit shifts across N.

At first glance, such drift could be misinterpreted as inconsistent feature usage—or even as evidence that the feature is becoming "rusty" or unimportant. However, as we show in Section 3.5, this interpretation is misleading. Although the *center* of the logit distribution is highly mobile across N, the feature's *discriminative margin* (i.e., the label-gap signal) remains strong and nearly invariant. This indicates that the feature continues to participate reliably in the classification mechanism, but its baseline is re-centered in a prompt-dependent manner.

While mechanistically interesting, adaptive re-centering also introduces a practical pitfall for SAE-based analysis. Because the mean logit can shift across N—and can even become negative for some configurations—the corresponding post-nonlinearity activation (e.g., after ReLU/JumpReLU) may be driven to zero. As a result, purely activation-centric analyses (the de facto standard in much of the SAE literature) can systematically miss decision-relevant antipodal features, yielding an incomplete picture of the underlying circuit. This motivates an important caveat for SAE-based interpretability: *"for a subset of features, analyzing the full pre-nonlinearity logit signal can be substantially more faithful and informative than relying solely on nonnegative activations."* Given how widespread activation-only analyses are in prior work, we emphasize this as a critical consideration when studying classification mechanisms with SAEs.



(a) Amplitude (b) Normalized Total Variation (c) Contextual Adaptivity

Figure 6: **Distinct Characteristics of Antipodal Features.** (a) Antipodal features exhibit substantially higher average amplitude. We hypothesize that, because these features participate as bidirectional latent axes, the model traverses them more freely in both directions, enabling them to capture richer, more fine-grained semantic variation. (b) Antipodal features also show markedly lower normalized total variation (NTV), indicating reduced sensitivity to short-timescale, token-level fluctuations and greater robustness to local lexical noise. (c) Finally, we observe that antipodal feature logits can shift dramatically as the number of few-shot demonstrations (FSDs) varies. This highlights an important caveat for SAE-based analyses: focusing solely on nonnegative activations (e.g., post-ReLU) can obscure substantial, decision-relevant logit dynamics, particularly for antipodal features.